



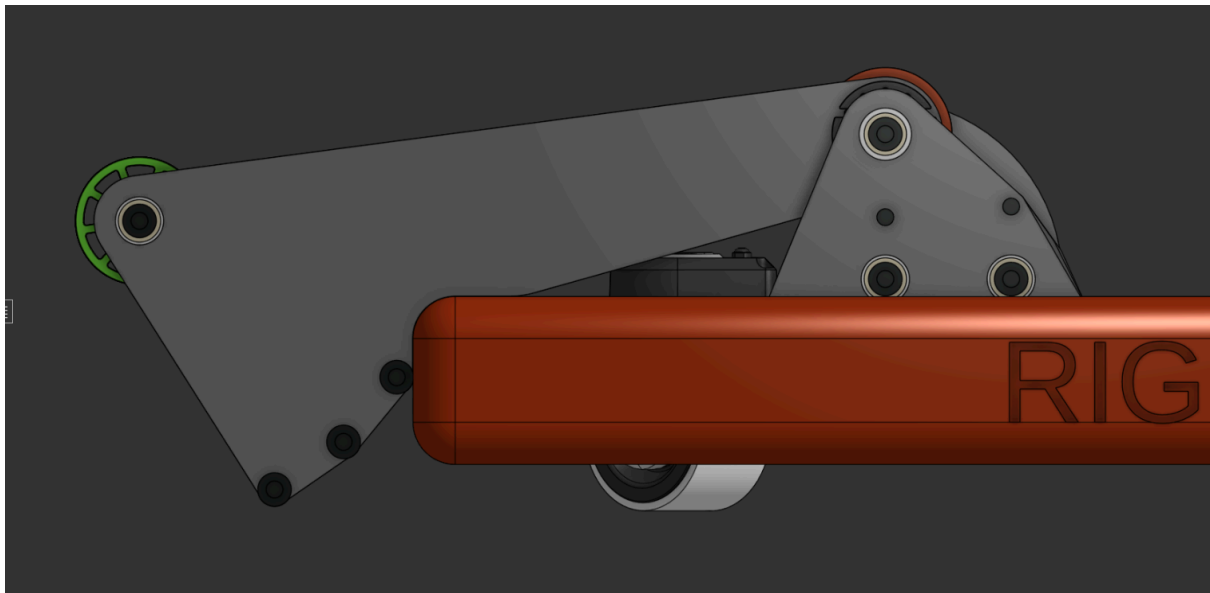
11x - Notes and Theoretical Background

Theoretical Background:

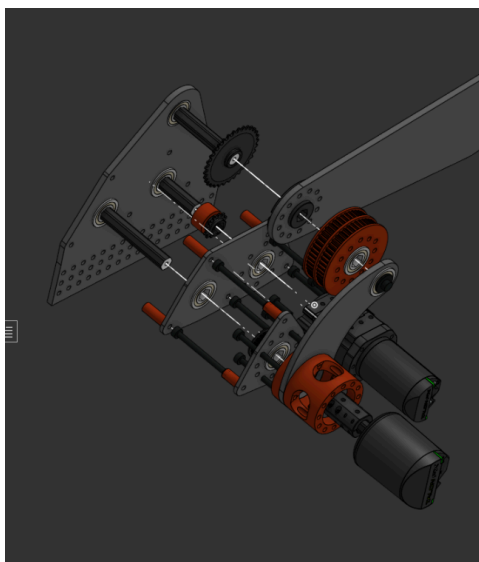
Before we get to the slapdown intakes we first need to understand what is an intake:

An intake is a main system in almost every year's robots. Its purpose is to acquire game pieces from outside the robot and get them into the robot perimeter and to the rest of the systems. **Basically, an intake is the first thing that meets the game piece during the match.**

A "slapdown" intake is a general name for a small family of intake designs that revolves around, well, revolving around something - in this case a rotating axis. The basic concept of a slapdown intake is constructed of two assemblies - the "static assembly" and the "rotating assembly". While the static assembly stays fully inside the robot perimeter at all times, even when the intake is deployed, the rotating assembly starts the game inside the robot perimeter and can rotate ("slap") all the way to outside the robot perimeter to deploy the intake.

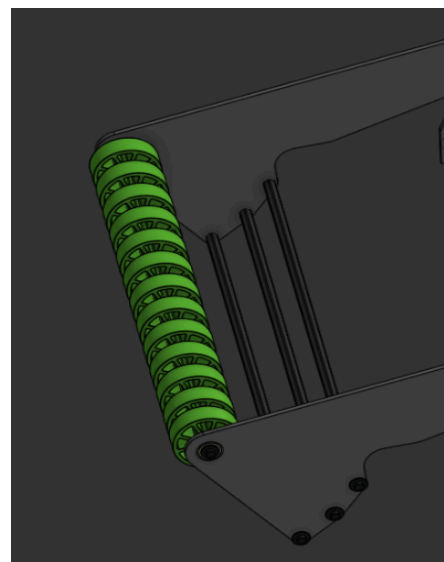


Intake deployed (mid match)



Static
assembly

Rotating
assembly



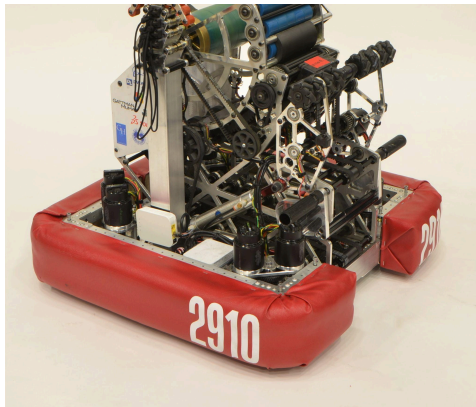


Cut Bumper Intake:

A cut bumper intake is possible only in years when the gap between the bumpers is larger than the game piece, that is because a cut bumper intake works by **passing the game piece through an opening in the bumper rather than above it**. In this file we will talk both on **Revolving** Cut Bumper Intake and **Static** Cut Bumper Intake.

Revolving:

The path the game piece takes in a RCBI² is the only major difference between it and an OTBI, as **they are both a slapdown intake by definition** (they both have static and rotating assemblies), therefore the difference in designing a RCBI is only in the path. While the intake geometry stays the same, the bumper constraint is now gone and is replaced by the inside of the robot. This means that while the bumper itself is not gone, it is no longer obstructing the path.

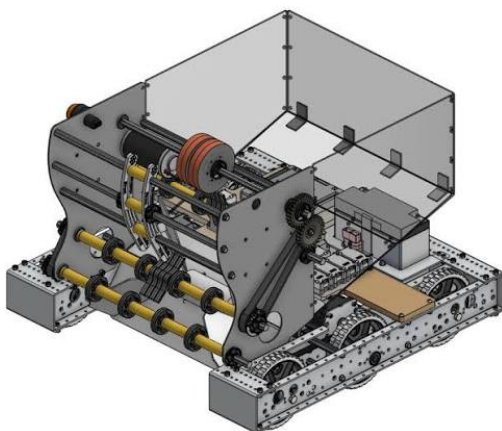


An example of a revolving cut bumper intake - 2910 in 2021.

Notice that there is still a slapdown intake above the cut part.

Static:

A SCBI³ has no revolve axis and is all part of the same static assembly, as part of that the entirety of the intake stays inside the robot perimeter at all times. Only being one static assembly means that the SCBI is not a “slapdown” intake. The geometry design for a SCBI includes only the path of the game piece and the desired compression.



An SCBI - KitBot 2026

2 - Revolving Cut Bumper Intake

3 - Static Cut Bumper Intake



Notes for presentation:

Slide 3:

What is a Slapdown Intake?

A slapdown intake is a pivoting mechanism commonly used in competitive robotics, particularly in FRC. It features an intake arm mounted to a pivot point inside the robot's frame. When activated, it quickly "slaps" or extends downward and outward to collect game pieces, and then retracts safely back inside the robot's bumper perimeter to avoid damage from collisions.

What is it Made Of?

A slapdown intake generally consists of two main sections:

- The Static Assembly: This includes the mounting frame, motors, and the main pivot axle. It remains completely inside the robot's perimeter at all times.
- The Rotating Intake: This includes the rollers, wheels, and arms that grab the game pieces. It is connected to the static axle and is the part that moves during operation.

How Does it Work?

- Stored State: When inactive, the rotating intake is tucked completely inside the robot's frame or bumper perimeter to stay protected.
- Deployment: When the driver or autonomous program needs to collect an object, the motors actuate the pivot axle, causing the intake arm to swing down and out past the bumpers.
- Retraction: Once the game piece is secured, the mechanism quickly pivots backward, bringing the intake arm safely back inside the robot.

Slide 4:

There are two main things to take into consideration when modeling a slapdown intake: The geometry of the intake itself and the constraints of the chassi. The latter is somewhat more complicated.

Checking the constraints:

The first step is to create a sketch with all the robot (chassi) constraints such as bumpers, electronics plate height, vertical and horizontal robot extension limits etc.

Defining the geometry:

The geometry changes every year and is dependent on that year's game piece. For example, let's take the 2026's gp – the fuel. On the right picture you can see three elements: The fuel (the big circles) with a diameter of 150mm, a 3in compliant wheel (top left) and a 0.5in kicker bar (bottom middle).

At this point you can start to define the compression on the game piece and therefore the placement of the wheels and game pieces in the intake.

Defining the path:

Defining the path is a continuation step of defining the geometry where you model the expected path the game piece is going to go through and derive from it the final placement of every part of the system.

The difference with cut bumper:

The only difference is that now the bumper is no longer an obstruction to the path of the game piece and can be ignored.

Slide 5:

A slapdown intake assembly consists of many parts, both in the static and the rotating sub-assemblies.

The main parts are the motors that drive everything, to them are connected the pulleys, gears and sprockets that transmit the rotation to the wheel, and of course the wheels themselves - all held by the plates.

A complete slapdown usually consists of 4 to 6 plates: a pair of identical plates to hold the rotating assembly parts and the other plates to hold the motor and the static assembly.



Slide 6:

After you have all the plates designed it's finally time for the assembly!

During the design process we thought about motor placement and integrated it in our plates, now we will place the actual motors in their place, create spacers and insert COTS. While there is no "correct" way to place everything, you should always strive to design your systems as efficiently as you can - space wise, weight wise and price wise.

The price of a system is mainly dependent on the amount and type of COTS used - for example, bearing will add to the price much less than motors. But choosing motors can be tough - what motor to use? With gearbox? And if so what ratio? Generally, the best way to choose motors is to define exactly how we want the intake to work and then calculate the minimum requirements needed to achieve them and take something that fits with extras, but it is also good to just design all the motors so you could change their gearbox by trial and error.

Example for calculations*: Let's say the rotating assembly off our intake weighs 2kg and is extending up to 30cm from the pivot point. We could say the intake is "perfect" and its center of mass will be exactly halfway to the extension (15cm). Using the law of leverage to calculate how much power is needed to lift the entire rotating assembly and checking motor data sheets we can find which motor can lift everything with extra tolerance, with good speed (rpm) while consuming moderate power.

*Such calculations are quite advanced and may take your team some time to accomplish so I generally do not suggest newer or less experienced teams use this method, and use the second method of "trial and error".